Tables Break Dense Retrieval: BM25, Hybrid Fusion, and an Error Taxonomy on Text-and-Table Documents

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Abstract

Retrieval-augmented generation is bounded by retrieval, but standard retrieval benchmarks (MTEB, BEIR) evaluate on homogeneous text where dense encoders typically win. Whether that ordering survives on documents combining prose and tables is an open empirical question. We evaluate nine retrieval configurations on T^2-RAGBench (23,088 financial QA queries over 7,318 SEC-filing documents averaging $\sim$920 tokens), together with a BGE-M3 robustness panel that varies the encoder and a ColBERTv2 follow-up driven by our error analysis.

Main findings. (1) BM25 beats `text-embedding-3-large` on every metric except Recall@20: MTEB and BEIR rankings do not transfer to text-and-table corpora. (2) Hybrid RRF + cross-encoder reranking reaches R@5 = 0.816 with closed APIs (Cohere) and R@5 = 0.751 with a fully open BGE-M3 stack (BGE-M3 dense encoder + bge-reranker-v2-m3, both at their fine-tune context lengths); the closed/open gap is 6.5 pp, with hybrid fusion absorbing most of the encoder-side difference (Section 6). (3) Of the 30.5% of queries hybrid RRF misses at rank 5, 73% [Wilson 95% CI: 64–81%, $n = 100$] fail because the answer sits in a table cell that linearized-text encoders cannot recover; ColBERTv2 lifts dense retrieval substantially but does not absorb this failure mode. (4) HyDE and multi-query expansion hurt on numerical queries (hallucinated figures pull the query embedding off-target), while a cross-encoder reranker on a candidate pool of ≥ 50 is the single largest improvement we measured (+12.1 pp R@5 over Hybrid RRF). All comparisons use paired bootstrap tests with Bonferroni correction. We release evaluation code, prompts, FAISS indices, and per-query outputs.

Keywords: retrieval-augmented generation; retrieval evaluation; hybrid retrieval; cross-encoder reranking; text-and-table documents; financial QA; error taxonomy.

1 Introduction

In a RAG system [Lewis et al., 2020], a retriever supplies the context an LLM conditions on. If the retriever misses the relevant document, no amount of prompt engineering recovers the answer: retrieval sets a hard upper bound on the whole pipeline. Most RAG research nevertheless treats retrieval as a solved substrate and focuses attention on prompting, decoding, and judging.

The gap for documents combining text and tables. That assumption breaks down on documents that mix prose and tables. Financial filings, earnings reports, and regulatory documents are built this way: the prose carries context, the tables carry the numbers. Answering *"What was the 2019 year-over-year revenue growth?"* requires finding both the right document and the right cell inside it. The

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benchmarks most practitioners consult, MTEB [Muennighoff et al., 2023] and BEIR [Thakur et al., 2021], evaluate on homogeneous text, where dense retrieval usually outperforms sparse baselines. Whether that ordering survives a shift to heterogeneous documents is an open empirical question, and our main finding is that it does not.

We use T^2-RAGBench [Strich et al., 2026], a financial QA benchmark of 23,088 queries over 7,318 documents. The original paper evaluated six retrieval methods with two metrics; we evaluate nine retrieval configurations together with a BGE-M3 robustness panel that varies encoder choice and a ColBERTv2 follow-up driven by our error analysis, and report the full retrieval and generation metric suite along with paired bootstrap significance tests.

Contributions.

1. A head-to-head evaluation of nine retrieval configurations on T^2-RAGBench, covering sparse, dense, hybrid, cross-encoder-reranked, query-side, index-side, and adaptive pipelines, with paired bootstrap significance tests, together with a BGE-M3 robustness panel that varies encoder choice and a ColBERTv2 follow-up evaluating late-interaction retrieval against the error taxonomy.
2. Evidence that rankings on MTEB and BEIR do not transfer to documents combining text and tables: BM25 beats a strong dense encoder (`text-embedding-3-large`) on nearly every metric.
3. An error analysis of 100 sampled retrieval failures showing that 73% share a single cause: the answer is a table cell that linearized-text encoders cannot locate. We argue this is the next concrete open problem in the area.
4. Two negative results: HyDE and multi-query expansion degrade retrieval on numerical financial queries, and CRAG-style correction does not close the gap to hybrid fusion.
5. Ablations on fusion type and reranker candidate depth, plus a released evaluation framework with prompts, indices, and per-query outputs.

2 Related Work

RAG retrieval methods. BM25 remains competitive in zero-shot settings [Thakur et al., 2021]. Dense dual-encoders [Karpukhin et al., 2020] and their successors (E5 [Wang et al., 2022], E5-Mistral [Wang et al., 2024], BGE-M3 [Chen et al., 2024a]) advance MTEB rankings through contrastive pretraining. Late-interaction retrievers (ColBERT [Khattab and Zaharia, 2020], ColBERTv2 [Santhanam et al., 2022]) keep a per-token vector for each document position rather than pooling, trading index size for finer-grained matching. Hybrid retrieval fuses sparse and dense signals via RRF [Cormack et al., 2009], consistently improving recall by 15–30% [Li et al., 2025]. Two-stage reranking with cross-encoders further refines results [Gao et al., 2024]. Query-side methods include HyDE [Gao et al., 2023] and RAG-Fusion [Raudaschl, 2024]. Index-side methods include Contextual Retrieval [Anthropic, 2024] and HyPE [Vake et al., 2025]. Adaptive approaches like CRAG [Yan et al., 2024] trigger corrective searches when confidence is low.

QA over documents that mix prose and tables. HybridQA [Chen et al., 2020] introduced multi-hop reasoning over linked tables and passages. In the financial domain, FinQA [Chen et al., 2021], TAT-QA [Zhu et al., 2021], and ConvFinQA [Chen et al., 2022] target numerical reasoning over hybrid contexts. A recent survey [Nan et al., 2025] finds that retrieval of heterogeneous context remains the primary bottleneck.

RAG benchmarks. BEIR [Thakur et al., 2021] established zero-shot retrieval evaluation across 18 datasets. Recent RAG benchmarks include RGB [Chen et al., 2024b], CRAG [Yang et al., 2024], and RAGBench [Friel et al., 2024]. Li et al. [2025] study RAG design choices at COLING 2025. None focus on retrieval method comparison over documents that mix prose and tables. T^2-RAGBench [Strich et al., 2026] unifies FinQA, ConvFinQA, and TAT-DQA into 23,088 queries over 7,318 financial documents; where the original paper tested six methods with two metrics, we evaluate nine retrieval configurations together with a BGE-M3 robustness panel and a ColBERTv2 follow-up, with a comprehensive set of retrieval and generation metrics.

3 Methodology

3.1 Dataset

T²-RAGBench [Strich et al., 2026] contains 23,088 question-context-answer triples from FinQA (8,281), ConvFinQA (3,458), and TAT-DQA (11,349), covering 7,318 unique financial documents averaging ∼920 tokens each. Each document mixes text and markdown-formatted tables from SEC filings. The benchmark reformulated all questions using Llama 3.3-70B to incorporate identifying information (company name, sector, report year), producing questions with exactly one correct answer regardless of context. All answers are numerical. The original best method reached only 41% Number Match against an oracle ceiling of 72–79%.

3.2 Retrieval Methods

We evaluate nine retrieval configurations spanning five categories, together with three BGE-M3 configurations that vary the encoder and a ColBERTv2 follow-up. Full hyperparameters are in Appendix A; prompt templates in Appendix C.

BM25. Okapi BM25 [Robertson and Walker, 1994] with $k_1 = 1.2$, $b = 0.75$, providing strong lexical matching for domain-specific terminology.

Dense retrieval. We encode queries and documents with **OpenAI text-embedding-3-large** (3,072d) via Azure AI Foundry, indexing with FAISS `IndexFlatIP` for exact inner-product search.

Hybrid RRF. Fuses BM25 and dense ranked lists via Reciprocal Rank Fusion [Cormack et al., 2009] with $k = 60$.

Hybrid + Cohere Rerank. Two-stage pipeline: hybrid RRF retrieves 50 candidates, reranked by Cohere Rerank v4.0 Pro [Cohere, 2025], returning top 10.

HyDE. Generates a hypothetical answer passage via GPT-4.1-mini and retrieves with its embedding [Gao et al., 2023].

Multi-Query. Generates three query variants via GPT-4.1-mini, retrieves for each, and merges via RRF [Raudaschl, 2024].

Contextual Retrieval. Prepends LLM-generated context summaries to each document at indexing time [Anthropic, 2024]. We apply it to both the dense and hybrid pipelines, yielding two configurations (*Contextual Dense*, *Contextual Hybrid*) reported separately. With seven other methods listed above, this brings the count to nine retrieval configurations.

CRAG. Classifies retrieved documents as relevant/ambiguous/irrelevant; rewrites query and re-retrieves when confidence is low [Yan et al., 2024].

BGE-M3 robustness panel. To test whether the encoder choice changes the ranking, we evaluate three additional configurations built on the BGE-M3 dense encoder (1024-d) [Chen et al., 2024a] and the BAAI bge-reranker-v2-m3 cross-encoder: dense retrieval, hybrid RRF (BM25 + BGE-M3), and the corresponding two-stage Hybrid + BGE Rerank pipeline. Both rerankers run inside their fine-tune context windows; configuration details are in Section 3.3.

ColBERTv2 (late interaction). As a follow-up to the table-structure failure mode identified by our error taxonomy (Section 5), we additionally evaluate ColBERTv2 [Santhanam et al., 2022] as a single-method retriever. ColBERTv2 retains a per-token vector for every document position rather than pooling to a single embedding, so a table cell whose label and value sit in different cells of the same row can still match a query token-by-token. We use the official `colbert-ir/colbertv2.0` checkpoint via `ragatouille` with `max_document_length`=512 (the value used during ColBERTv2 fine-tuning); documents longer than 512 tokens are split into passages, scored independently, and aggregated to document level by max passage score per query.

Category	Method	R@1	R@3	R@5	R@10	MRR@3	nDCG@10	MAP
Single-method	BM25 (sparse)	0.293	0.552	0.644	0.735	0.411	0.515	0.449
	Dense (text-embed-3-large)	0.248	0.481	0.587	0.703	0.351	0.466	0.398
	Dense (BGE-M3)$^\diamond$	0.229	0.434	0.515	0.607	0.321	0.413	0.357
	ColBERTv2 (late interaction)$^\diamond$	0.301	0.526	0.616	0.709	0.402	0.501	0.440
Query expansion	HyDE (gpt-4.1-mini)	0.221	0.441	0.544	0.671	0.318	0.433	0.365
	Multi-Query + RRF	0.283	0.539	0.640	0.734	0.397	0.506	0.439
Index augment.	Contextual Dense	0.266	0.508	0.615	0.732	0.373	0.490	0.420
	Contextual Hybrid	0.327	0.610	0.717	0.818	0.454	0.571	0.497
Adaptive	CRAG (gpt-4.1-mini)	0.302	0.556	0.658	0.788	0.415	0.536	0.456
Fusion + rerank	Hybrid (BM25+Dense, RRF)	0.308	0.588	0.695	0.801	0.433	0.551	0.477
	Hybrid (BGE-M3, RRF)$^\diamond$	0.290	0.545	0.641	0.741	0.404	0.513	0.445
	Hybrid + Cohere Rerank	**0.472**	**0.758**	**0.816**	**0.861**	**0.605**	**0.683**	**0.625**
	Hybrid (BGE-M3) + BGE Rerank$^\diamond$	0.383	0.674	0.751	0.822	0.516	0.612	0.547

Table 1: Main retrieval results on T^2-RAGBench (23,088 queries, 7,318 documents). $^\diamond$Open-weight rows run locally on a single GPU: BGE-M3 (`BAAI/bge-m3`, 1024-d) and `BAAI/bge-reranker-v2-m3` for the BGE-M3 stack, and ColBERTv2 (`colbert-ir/colbertv2.0`, late interaction) as a separate single-method retriever. Unmarked rows use Azure AI Foundry endpoints (`text-embedding-3-large`, Cohere Rerank). Latency is omitted because cross-hardware comparison would be misleading. All within-stack pairwise differences are statistically significant at $p < 0.05$ after Bonferroni correction over the comparison family, with the exceptions noted in Section 4.2. Best in **bold**.

3.3 Evaluation

Retrieval: Recall@k ($k \in \{1, 3, 5, 10, 20\}$), MRR@$k$, nDCG@$k$, MAP. **Generation:** Number Match (NM) with tolerance $\epsilon = 10^{-2}$. **Statistics:** Paired bootstrap tests ($B = 10{,}000$, Bonferroni-corrected, $p < 0.05$).

Setup. BM25 and FAISS run locally on Apple Silicon. Embedding, LLM calls, and Cohere reranking use Azure AI Foundry. The BGE-M3 robustness panel uses `BAAI/bge-m3` (1024-d) [Chen et al., 2024a] and the `BAAI/bge-reranker-v2-m3` cross-encoder, both running locally on a single NVIDIA RTX PRO 6000 Blackwell GPU (102 GB). Each model is configured at its officially recommended operating point: dense encoders at 8,192-token context (BGE-M3's native maximum and approximately the OpenAI API limit), Cohere Rerank v4.0 Pro at `max_tokens_per_doc`=4,096 (default), and bge-reranker-v2-m3 at `max_length`=1,024 (BAAI's stated fine-tuning length; the 512 used in the model-card example is a demo setting and truncates the 920-token mean T^2-RAGBench document). Each candidate document is clipped to the first 4,096 characters before reranking. Candidate pools are 50 for Cohere and 60 for BGE, both above the 20-candidate threshold below which reranker quality degrades sharply (Section 4.4). Both rerankers run inside their trained context windows; we do not extend either beyond its fine-tune length. Latency is reported only within hardware-matched comparisons. All experiments use seed 42, temperature 0, and the standard train/test split. Code is publicly released [Anonymous Authors, 2026].

4 Results

4.1 Main Retrieval Results

Table 1 presents retrieval performance of all evaluated methods on the full T^2-RAGBench test set (23,088 queries, 7,318 documents).

Hybrid RRF with a cross-encoder reranker is the clear winner: Recall@5 of 0.816, compared to 0.695 for unreranked hybrid (+17%), 0.644 for BM25 (+27%), and 0.587 for dense (+39%). MRR@3 moves from 0.433 to 0.605 on the same transition, a 40% relative gain. Among the first-stage retrievers, BM25 beats dense on every metric except Recall@20, where they are effectively tied. Contextual retrieval adds 2–3 pp of Recall@5 to both dense and hybrid for an LLM call per

Retrieval	**Number Match**
BM25	0.251
Dense	0.257
Hybrid RRF	0.282
Oracle	0.350

Table 2: End-to-end Number Match by retrieval method, GPT-4.1-mini generator, $\epsilon = 10^{-2}$ tolerance. Pearson $r = 0.980$ across these four points (Figure 6); BM25 and Dense are within 0.6 pp on NM, the only rank inversion versus Recall@5. The Oracle ceiling indicates retrieval headroom remains.

document at indexing time. CRAG edges past BM25 (+1.4 pp Recall@5) but not past hybrid fusion. Per-subset numbers are in Appendix A.

4.2 Negative Results: What Doesn't Work and Why

HyDE is counterproductive. HyDE underperforms vanilla dense retrieval (Recall@5: 0.544 vs. 0.587, $p < 0.001$), reproducing the result in Strich et al. [2026]. The failure is specific to numerical queries. Asked to draft a hypothetical answer for "What was the 2019 net revenue?", the LLM produces plausible-looking but fabricated figures; the resulting embedding is close to documents that happen to mention those invented numbers, not to the one with the true answer. HyDE's premise is paraphrase matching, and exact numerical answers are the setting where paraphrase does the least work.

Multi-query provides negligible benefit. Multi-query achieves Recall@5 of 0.640, statistically indistinguishable from BM25's 0.644 ($p = 0.31$). Financial queries are already specific; reformulations do not surface new documents because the retrieval gap is structural (table format), not lexical.

CRAG cannot match hybrid fusion. CRAG triggers a query-rewrite branch whenever its relevance evaluator marks the top retrieval as ambiguous or irrelevant; the rewritten queries face the same table-structure challenge as the originals. Query rewriting alone cannot substitute for combining complementary retrieval signals.

4.3 End-to-End Generation

Number Match tracks retrieval quality closely across the four retrievers (Table 2): BM25 and Dense are within 0.6 pp NM (Dense 0.257 above BM25 0.251, the one rank inversion versus Recall@5 in this panel), Hybrid RRF reaches 0.282, and the Oracle ceiling is 0.350. Pearson correlation against Recall@5 across the four points is $r = 0.980$ (Figure 6); the inversion is at the noise floor and the linear trend is otherwise tight. The Oracle–Hybrid gap of 6.8 pp is the room left after retrieval is fixed; everything below is retrieval headroom. A linear extrapolation from the four-point fit projects NM ≈ 0.31 at the Hybrid + Cohere Rerank operating point (R@5 = 0.816); we mark this as an extrapolation rather than a measurement, since end-to-end NM was evaluated only at the four points above.

4.4 Ablation Studies

Fusion method. Convex Combination with $\alpha = 0.5$ (equal weight on BM25 and dense) reaches Recall@5 of 0.726, beating RRF ($k = 60$) at 0.695; within RRF, $k = 10$ is the best setting we tried (0.716). Equal weighting wins in both cases (Figure 1).

Reranker depth. The dependence on candidate-pool size is a cliff, not a smooth ramp. At 20 candidates, hybrid RRF already places the gold document in the input pool 87.7% of the time (R@20, Table 5), but the reranker recovers only 52% of those: R@5 = 0.458. At 50 candidates the rerank R@5 jumps to 0.826, and at 100 it reaches 0.888 (Figure 2); the conversion rate rises sharply as

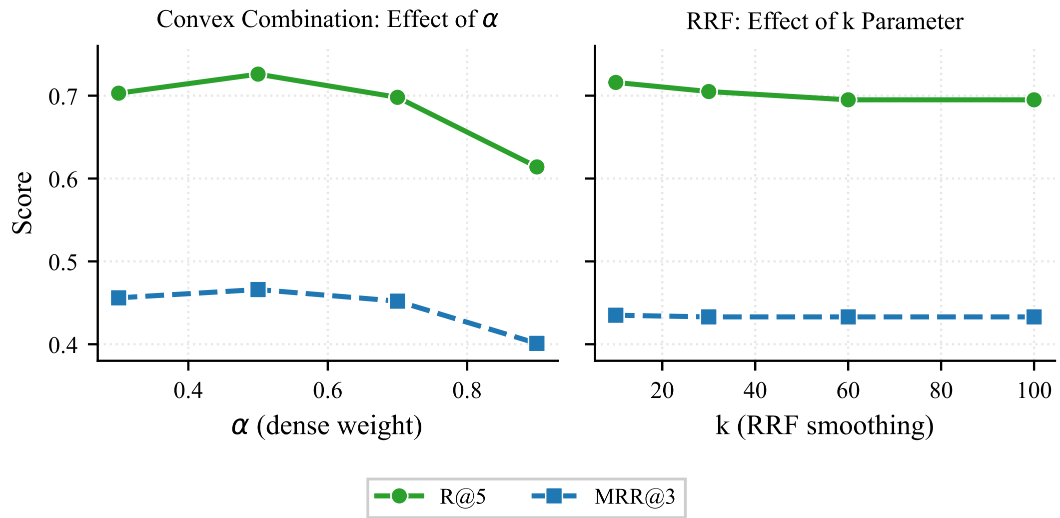

Figure 1: Fusion method ablation. Left: Convex Combination with varying α; $\alpha = 0.5$ is optimal. Right: RRF with varying k; lower k yields better results.

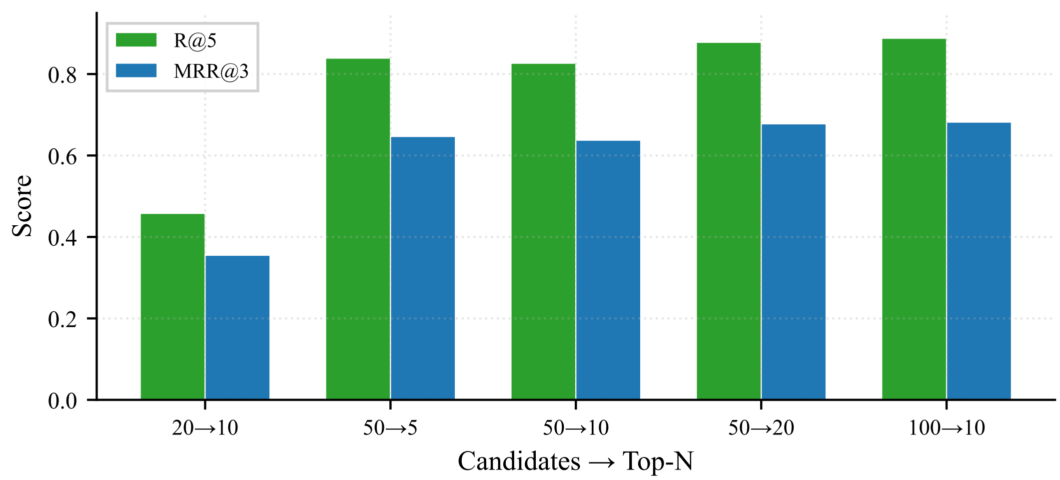

Figure 2: Reranker depth ablation. Critical threshold at 50 candidates.

the pool fills out, with diminishing returns past 100.[1] We conjecture, without testing it directly, that cross-encoders sharpen on candidate sets that span both clear positives and clear negatives, and that a short pool of all-relevant hybrid top-20 leaves the score margins between candidates noise-dominated. Practically, a candidate pool of at least 50 is the operating point we recommend.[2]

[1] Hybrid R@50 and R@100 were not part of our metric sweep, so the exact conversion rates at the larger pool sizes are bounded above (by the same R@N that they would denominate) but not directly measured. The 52% at pool=20 is exact: 0.458/0.877.

[2] The 0.826 reported here is from the reranker-depth ablation; Table 1 reports 0.816 for the same configuration from an independent run of the same setup. Cohere Rerank is not bit-deterministic in our pipeline even at temperature 0, and the 1 pp gap between these two runs is the in-paper variance estimate; we treat it as the floor for any single-run comparison against the Cohere stack.

Failure Category	%
Table structure mismatch	73
Numerical reasoning	20
Vocabulary mismatch	5
Ambiguous query	1
Long document	1

Table 3: Error taxonomy (n=100 sampled failures). Table structure mismatch dominates; Wilson 95% CI for the 73% headline rate is $[64\%, 81\%]$, and the category remains the modal failure across the entire interval.

5 Error Taxonomy for Text-and-Table Retrieval

The 30.5% of queries where hybrid RRF misses the gold document in its top-5 is not a uniform error surface. We sampled 100 of those 7,033 failures and had GPT-5.4 label each by failure type;[3] the resulting distribution (Table 3) is dominated by a single category.

Table structure mismatch (73%). Standard encoders see a table as linearized text; the two-dimensional row/column geometry is gone by the time the embedding is computed. A query like "What was net income in 2019?" cannot match a row whose label ("net income") and fiscal year ("2019") sit in different cells of the same row. This is the open problem the benchmark surfaces: table-aware embeddings or retrievers that preserve row/column structure.

Numerical reasoning (20%) failures occur when questions require computation (e.g., year-over-year change) rather than direct lookup.

Cross-subset patterns. TAT-DQA has the highest failure rate (35.3% vs. 26.3% FinQA, 24.6% ConvFinQA). Among failures, 71.0% of gold documents appear in *neither* the dense nor BM25 top-5, indicating genuinely hard cases rather than fusion artifacts.

Implications for evaluation. A method that pushes aggregate Recall@5 up might only be chipping at the vocabulary mismatch tail (5% of errors) while leaving the table structure mass (73%) untouched. Future evaluations on heterogeneous corpora should split metrics by failure category; the aggregate alone is misleading.

6 Discussion

Why BM25 wins, and what that means for leaderboards. `text-embedding-3-large` ranks among the top dense encoders on MTEB and BEIR, but on T^2-RAGBench BM25 beats it on every metric except Recall@20, where the two are tied. Financial queries are full of terms that appear verbatim in answer documents (company names, tickers, standardized metric labels, fiscal period identifiers), and lexical matching exploits this directly. A dense encoder distributes the same tokens across a neighborhood of nearby vectors, many of which the query never asked for. The lesson is narrower than "rethink evaluation": MTEB and BEIR rankings do not transfer to corpora that combine text and tables, and a dense encoder picked from those leaderboards is the wrong default for such data.

What changes when we swap to a fully open stack. Swapping `text-embedding-3-large` for the BGE-M3 dense encoder (1024-d) leaves the broad ranking intact but localises the gap. As a single-method retriever, BGE-M3 is significantly weaker than both OpenAI dense (R@5 = 0.515 vs. 0.587, $p < 0.001$) and BM25 (0.515 vs. 0.644, $p < 0.001$). Hybrid fusion absorbs the encoder difference: Hybrid (BM25 + BGE-M3, RRF) reaches R@5 = 0.641 against BM25 at 0.644, a paired-bootstrap difference of -0.31 pp ($p = 0.265$, not significant). The same fusion with OpenAI dense

[3]The 100-failure sample was originally drawn from a slightly earlier hybrid RRF run with R@5 = 0.689 (7,188 failures); the current run shifted R@5 to 0.695 and the failure set differs by $\sim$160 queries out of $\sim$7,000. Re-drawing on the current set is unlikely to materially change the categorical distribution; the 73% headline holds across the entire Wilson 95% CI [64%, 81%].

yields $+5.1$ pp over BM25 (0.695 vs. 0.644), so dense quality matters more for *amplifying* BM25 than for *matching* it. With both rerankers run at their fine-tune length, the two-stage results are closer than the encoder gap suggests. Cohere Rerank lifts hybrid by $+12.1$ pp on the OpenAI stack (0.695 $\rightarrow$ 0.816); bge-reranker-v2-m3 at `max_length`=1,024 lifts hybrid by $+11.0$ pp on the BGE-M3 stack (0.641 $\rightarrow$ 0.751, $p < 0.001$). The two rerankers move hybrid by nearly the same absolute amount; what remains is a 6.5 pp gap (0.816 vs. 0.751) between the closed and open pipelines, well below the per-subset variability we see in Appendix A. The 1,024 setting matters: at the model-card example value of 512, the open reranker instead *reduces* hybrid R@5 by 3.1 pp ($0.641 \rightarrow 0.610$, $p < 0.001$). On a corpus with 920-token mean document length the apparent regression at 512 is a truncation artifact, since BAAI's fine-tuning recipe uses 1,024 tokens; the difference between the two settings is a paired-bootstrap $+14.1$ pp ($p < 0.001$). We report the in-regime 1,024 number throughout and treat the 512 baseline as a truncation diagnostic, not a quality claim about the model.

Late-interaction retrieval is a partial fix. The error taxonomy points at table-cell answers as the dominant failure mode and flags late-interaction retrievers as the natural representational fix. We tested ColBERTv2 [Santhanam et al., 2022] as a single-method retriever, with passages encoded at `max_doc_length`=512 and aggregated to document-level by max passage score. ColBERTv2 lifts dense retrieval substantially (R@5 = 0.616 vs. 0.587 for `text-embedding-3-large`, $+2.9$ pp paired-bootstrap, $p < 0.001$; vs. 0.515 for BGE-M3, $+10.1$ pp, $p < 0.001$). It does not, however, beat BM25 (0.616 vs. 0.644, -2.8 pp, $p < 0.001$) and sits well below hybrid fusion (-8.0 pp vs. Hybrid RRF, $p < 0.001$). Per-subset, ColBERTv2 ties BM25 on ConvFinQA (0.695 vs. 0.696) but loses on FinQA and TAT-DQA, and notably does *not* disproportionately help on TAT-DQA, the most table-heavy subset. Token-level late interaction is a partial fix: it closes much of the gap from a dense encoder to BM25 without reaching it, and the table-structure failure mode is not absorbed by replacing soft attention over a pooled vector with max-similarity over linearized tokens.

The reranker does most of the heavy lifting. Adding a cross-encoder reranker on top of hybrid RRF is worth +17.2pp MRR@3 and +12.1pp Recall@5. That is a larger improvement than anything else we measured, and it is the only configuration that closes most of the gap to the oracle ceiling. Candidate depth matters: the rerank step degrades below 50 candidates (Section 4.4).

If you have to pick one setup. Use hybrid RRF with a cross-encoder reranker and a candidate pool of at least 50. Contextual retrieval is worth the one-time indexing cost: it lifts both dense and hybrid by roughly 2–3 pp of Recall@5 for an LLM call per document. Skip HyDE and multi-query expansion for numerical domains: HyDE actively moves the numbers in the wrong direction, and multi-query does not close the gap to BM25. Do not extrapolate from MTEB or BEIR; evaluate on data that looks like yours.

7 Conclusion

On a financial QA benchmark of 23,088 queries that mix prose and tables, hybrid RRF followed by a cross-encoder reranker delivers the best end-to-end performance; stacking query expansion on top of a dense encoder makes things worse. Two findings are worth carrying to other heterogeneous corpora. BM25 beats a state of the art dense encoder on nearly every metric, so MTEB and BEIR rankings should not be taken as guidance for data this isn't drawn from. And 73% of the retrieval failures that remain share a single cause: the answer is a table cell, and the encoder processes tables as linearized text. Closing that gap is less about scaling the encoder than about representing tabular structure. Token-level late interaction (ColBERTv2) is a partial fix — it lifts dense retrieval substantially but does not beat BM25 and does not solve the table-cell case. Promising directions we did not explore are structure-aware chunking, table-aware encoders such as Jina-ColBERT-v2 [Sturua et al., 2024] or TAPAS-style architectures, and using late-interaction inside hybrid fusion or as a reranker rather than as a single-method retriever.

Limitations

One domain. The corpus is financial filings. A different domain (legal briefs with tables, medical records, scientific papers) may shift both the absolute numbers and the ordering between methods, though we expect BM25's edge to be strongest where terminology is standardized and weakest where paraphrase is common.

Numerical answers only. Every answer in T^2-RAGBench is a number, which is why Number Match is the primary generation metric. We do not test whether the same retrieval ordering holds for free-form answers. End-to-end NM was measured at four operating points (Dense, BM25, Hybrid RRF, Oracle); the two-stage pipelines (Hybrid + Cohere Rerank, Hybrid (BGE-M3) + BGE Rerank) and the BGE-M3 single-method panel were evaluated on retrieval only. The linear projection in Section 4.3 is an extrapolation, not a measurement.

Whole-document retrieval. We index each document whole ($\sim$920 tokens). Chunking changes the recall game; it may close part of the dense encoder gap on longer documents by giving the encoder a smaller surface to summarize.

Two dense encoders, two rerankers. We evaluated two dense encoders (`text-embedding-3-large`, BGE-M3) and two cross-encoder rerankers (Cohere Rerank v4.0 Pro, `bge-reranker-v2-m3`). The qualitative ordering (BM25 above either dense encoder, hybrid above both, hybrid+rerank above hybrid) held in every pairing we tested once both rerankers ran inside their fine-tune context windows (4,096 and 1,024 tokens respectively). Encoder families with explicit table-aware pretraining may behave differently, and we do not claim universality beyond either reranker's trained regime.

Commercial APIs. The dense encoder, reranker, and generation LLM are all commercial endpoints. Model weights and behaviors can change. We release per-query outputs so that our results can be compared against future runs rather than re-reproduced from scratch.

Ethics Statement

The source data is public: SEC filings, accessed through the T^2-RAGBench release. No personal, private, or human-subject data is used, and all LLM calls go through commercial endpoints under their standard terms. The main positive externality we expect is cheaper, more accurate retrieval in RAG systems built over regulatory or financial documents. The main negative externality to flag is that better retrieval over SEC filings could make it easier to automate certain information asymmetry exploits (e.g., faster-than-human extraction of earnings figures for algorithmic trading). The mitigation is identical to the existing state of the art: these documents are already public, and practitioners can already search them. Releasing a stronger retrieval baseline does not change who has access; it changes only how fast the useful signal can be surfaced.

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Method	Parameter	Value	Notes
BM25	k_1 / b Tokenizer	1.2 / 0.75 whitespace split	Canonical Okapi defaults Via `rank_bm25`
Dense (text-embed-3-large)	Dimensions Max input Index type	3,072 $\sim$8,191 tokens FAISS `IndexFlatIP`	OpenAI via Azure AI Foundry API default Exact cosine search
Dense (BGE-M3)$^\diamond$	Dimensions `max_seq_length` Index type	1,024 8,192 FAISS `IndexFlatIP`	`BAAI/bge-m3` via `sentence-transformers` BGE-M3 native maximum Exact cosine search
Hybrid RRF	k	60 (ablation: 10, 30, 100)	Smoothing constant
Hybrid CC	α	0.5 (ablation: 0.3, 0.7, 0.9)	Dense weight
Cohere Rerank	Model / top_n `max_tokens_per_doc` Candidate pool	v4.0-pro / 10 4,096 50	Via Azure AI Foundry API default From first stage
BGE Rerank$^\diamond$	Model / top_n `max_length` Candidate pool	`bge-reranker-v2-m3` / 10 1,024 60	Local cross-encoder BAAI fine-tune length From first stage; minor asymmetry vs Cohere (50)
HyDE	LLM / temp / max_tok	gpt-4.1-mini / 0 / 150	Single generation
Multi-Query	Variants / RRF k	3 / 60	gpt-4.1-mini, temp 0
CRAG	Eval temp / rewrite temp	0.0 / 0.5	gpt-4.1-mini
Contextual	LLM / temp / max_tok	gpt-4.1-mini / 0 / 100	Per-document context
Global	Seed / top-k Bootstrap	42 / {1,3,5,10,20} B=10,000, p<0.05	All methods Bonferroni-corrected

Table 4: Complete hyperparameter settings for all retrieval methods.

Method	R@1	R@3	R@5	R@10	R@20	MRR@3	nDCG@10	MAP
Dense (BGE-M3)$^\diamond$	0.229	0.434	0.515	0.607	0.688	0.321	0.413	0.357
HyDE	0.221	0.441	0.544	0.671	0.767	0.318	0.433	0.365
Dense	0.248	0.481	0.587	0.703	0.798	0.351	0.466	0.398
Contextual Dense	0.266	0.508	0.615	0.732	0.817	0.373	0.490	0.420
ColBERTv2 (late interaction)$^\diamond$	0.301	0.526	0.616	0.709	0.780	0.402	0.501	0.440
Multi-Query	0.283	0.539	0.640	0.734	0.820	0.397	0.506	0.439
Hybrid (BGE-M3)$^\diamond$	0.290	0.545	0.641	0.741	0.819	0.404	0.513	0.445
BM25	0.293	0.552	0.644	0.735	0.797	0.411	0.515	0.449
CRAG	0.302	0.556	0.658	0.788	0.788†	0.415	0.536	0.456
Hybrid RRF	0.308	0.588	0.695	0.801	0.877	0.433	0.551	0.477
Contextual Hybrid	0.327	0.610	0.717	0.818	0.887	0.454	0.571	0.497
Hybrid (BGE-M3) + BGE Rerank$^\diamond$	0.383	0.674	0.751	0.822	0.867	0.516	0.612	0.547
Hybrid + Cohere Rerank	**0.472**	**0.758**	**0.816**	**0.861**	*	**0.605**	**0.683**	**0.625**

Table 5: Full retrieval results sorted by nDCG@10. Rows marked $^\diamond$ use open-weight retrievers run locally on a single GPU (BGE-M3 stack or ColBERTv2); unmarked rows use text-embedding-3-large via Azure. *Rerank returns ≤10 documents. †CRAG returns ≤10 documents; R@20 set equal to R@10 for completeness.

A Hyperparameter Details and Full Results

Table 4 lists all hyperparameters. Table 5 presents full retrieval metrics. Table 6 shows per-subset performance.

B Additional Figures

C Prompt Templates

All prompts use `gpt-4.1-mini` via Azure AI Foundry.

Subset	Method	R@5	R@10	MRR@3
FinQA	BM25	0.729	0.834	0.389
	Dense	0.611	0.748	0.308
	Hybrid	0.737	0.856	0.389
	Dense (BGE-M3)$^\diamond$	0.611	0.715	0.373
	Hybrid (BGE-M3) + BGE Rerank$^\diamond$	0.829	0.891	0.529
	Hybrid + Cohere Rerank	**0.860**	**0.909**	**0.578**
ConvFinQA	BM25	0.696	0.781	0.500
	Dense	0.654	0.781	0.410
	Hybrid	0.754	0.850	0.519
	Dense (BGE-M3)$^\diamond$	0.585	0.689	0.332
	Hybrid (BGE-M3) + BGE Rerank$^\diamond$	0.826	0.882	0.600
	Hybrid + Cohere Rerank	**0.874**	**0.907**	**0.671**
TAT-DQA	BM25	0.566	0.649	0.400
	Dense	0.549	0.647	0.364
	Hybrid	0.647	0.746	0.438
	Dense (BGE-M3)$^\diamond$	0.423	0.503	0.279
	Hybrid (BGE-M3) + BGE Rerank$^\diamond$	0.672	0.753	0.481
	Hybrid + Cohere Rerank	**0.766**	**0.812**	**0.605**

Table 6: Per-subset results. Hybrid + Cohere Rerank is the per-subset winner on every metric we report; among the two-stage open pipelines, BGE-M3 + BGE Rerank is consistently second and lifts the BGE-M3 hybrid base by 11–19 pp R@5 per subset. TAT-DQA is the hardest subset across the board. Rows marked $^\diamond$ use the BGE-M3 stack (`BAAI/bge-m3` + `bge-reranker-v2-m3`).

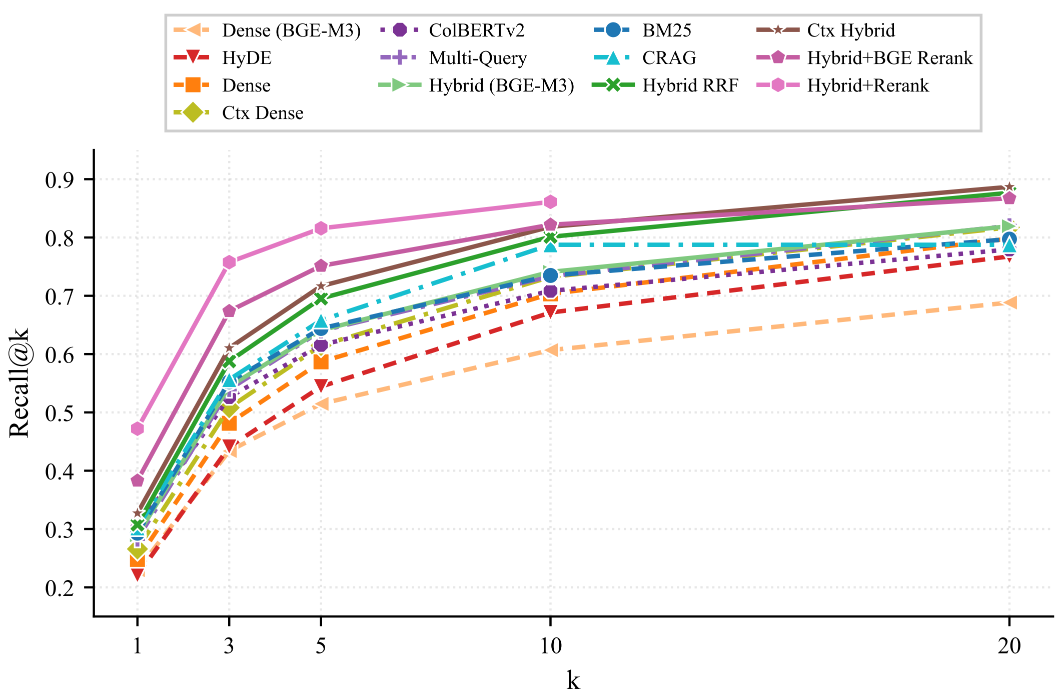

Figure 3: Recall@k curves. Hybrid fusion outperforms both baselines at every k, with the largest gains at small k.

C.1 Generation Prompt

```
Answer the following question based ONLY on
the provided context.
If the answer is a number, provide just the
number. If you cannot answer from the
```

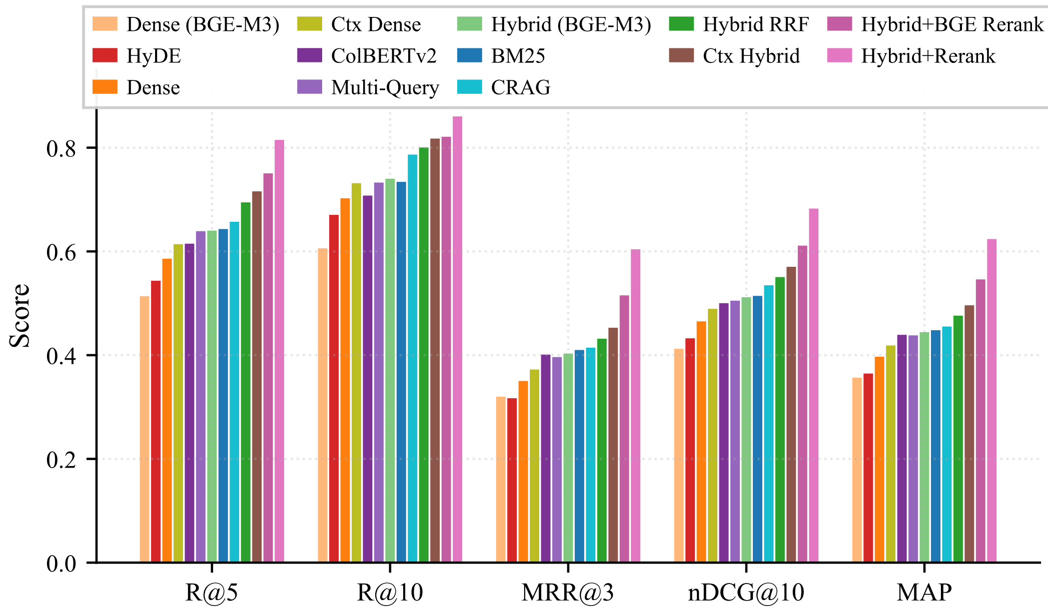

Figure 4: Grouped comparison across five metrics. BM25 outperforms standalone dense retrievers; hybrid + cross-encoder reranking is the top configuration on every metric.

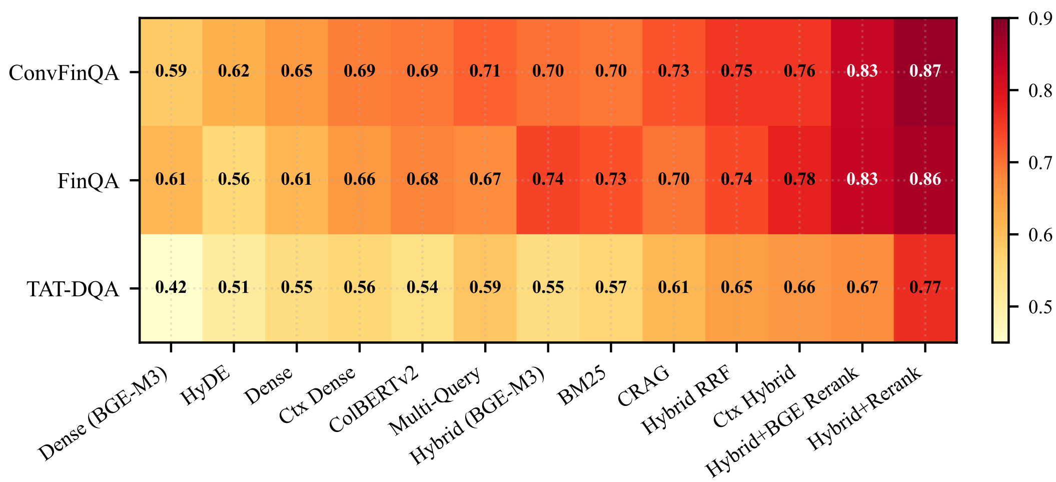

Figure 5: Recall@5 heatmap across methods and subsets. TAT-DQA is consistently hardest.

```
context, say "UNANSWERABLE".

Context: {context}
Question: {question}
Answer:
```

C.2 HyDE Prompt

```
Given the following question about financial
data, write a short passage that would
contain the answer. Include specific numbers
and financial terms.
Question: {query}
Passage:
```

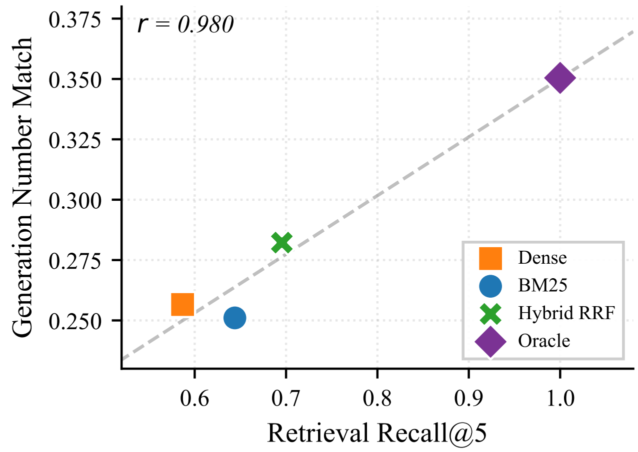

Figure 6: Retrieval and generation correlation ($r = 0.980$, $n = 4$): better retrieval tracks better answers. Four-point ablation (Dense / BM25 / Hybrid RRF / Oracle) on the GPT-4.1-mini generator. We did not measure NM end-to-end for Hybrid + Cohere Rerank or for the BGE-M3 stacks; the linear projection from this fit is reported in Section 4.3.

C.3 Multi-Query Prompt

```
You are a helpful assistant that generates
alternative search queries. Given the
following question, generate {n} alternative
phrasings that capture the same information
need but use different wording or
perspectives. Return each query on its own
line, numbered. Do not include any other text.

Original question: {query}
Alternative queries:
```

C.4 CRAG Evaluation Prompt

```
You are a relevance evaluator. Given a
question and a retrieved document, classify
the document's relevance.

Question: {query}
Document: {document}

Respond with exactly one of:
- RELEVANT
- AMBIGUOUS
- IRRELEVANT

Classification:
```

C.5 CRAG Rewrite Prompt

```
The following question was used to search a
financial document corpus, but the retrieved
results were not sufficiently relevant.

Original question: {query}

Please rewrite this question to be more
specific and likely to retrieve the correct
financial document.
```

```
Rewritten question:
```

C.6 Contextual Retrieval Prompt

```
Here is a document:
<document>
{document}
</document>

Please provide a concise summary context
(2-3 sentences) that captures the key
topics and entities in this document, for
the purpose of improving search retrieval.
Answer only with the context, nothing else.
```